Subject	**:**	**Artificial Intelligence (SECJ3553)**
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Theme	**:**	**Smart Education (SimuLearn)**
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Lecturer	**:**	**Dr. Norsham binti Idris**

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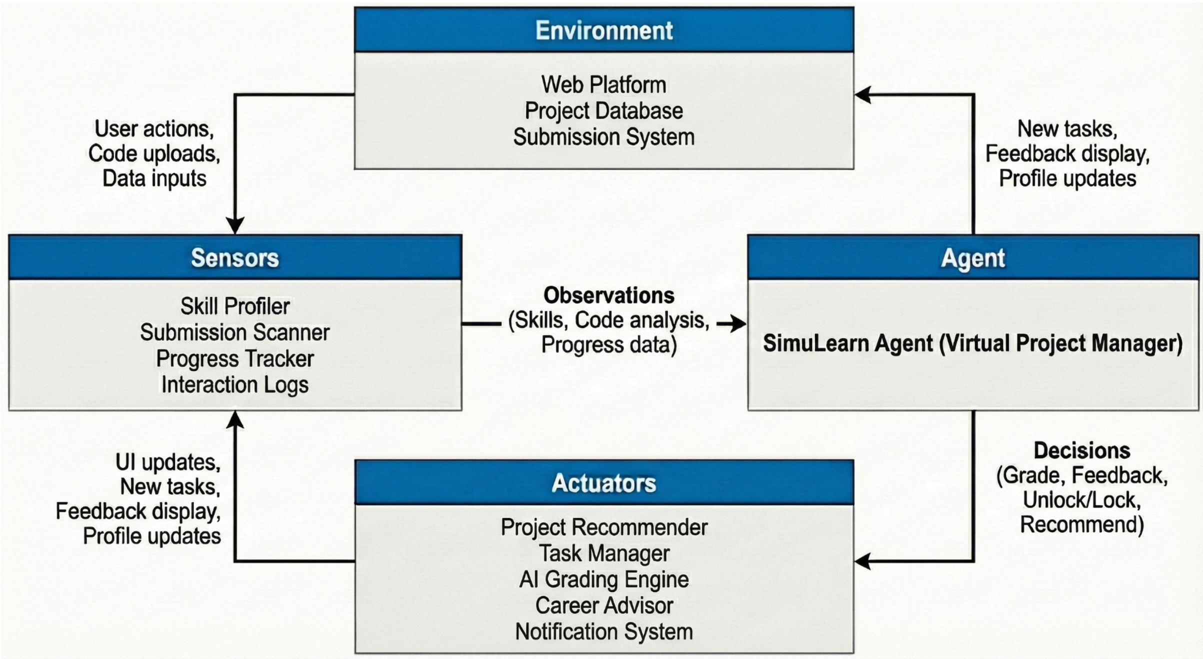
1.0 PEAS Model

SimuLearn acts as an intelligent Virtual Sensor Project Manager, designed to guide students through realistic industry simulations.

Property	Description
Performance	• **Acquire Student Skills:** Ensure students learn new technical skills. • **Grade Accurately:** Evaluate code when compared against predefined industry standards. • **Recommend Relevantly:** Suggest projects that match the student's likes and needs. • **Provide Quality Feedback:** Provide clear, and helpful advice after task or project submission. • **Engage Users:** Task and project completion rates, indicating sustained learner motivation.
Environment	• **Web Platform:** The SimuLearn dashboard, project library, skill profile, and task workflow interface. • **Project Database:** Repository of simulated real-world projects, task dependencies, constraints, and evaluation criteria. • **Submission System:** Interface that receives source code files, textual explanations, and user inputs for assessment.
Actuators	• **Project Recommender:** Suggests suitable projects based on inferred student skills and learning gaps. • **Task Manager:** Unlocks or blocks task nodes according to prerequisite satisfaction and learning progression rules. • **AI Grading Engine:** Automatically evaluates submissions, assigns scores, and generates structured feedback. • **Career Advisor:** Updates student skill profiles and provides personalized career path suggestions. • **Notification System:** Sends system messages such as pass or fail alerts.
Sensors	• **Skill Profiler:** Collects self-reported skills, syllabus topics, and free-text input from students.

	Submission Scanner: Analyzes uploaded code structure, syntax, and compliance with task requirements. **Progress Tracker:** Monitors task completion status, time-on-task, and retries. **Interaction Logs:** Tracks user interactions such as hints accessed, and resources viewed.

2.0 Relations Between Each Property



The diagram shows how each property interacts with each other.

1. **Environment -> Sensors:** The cycle begins in the **Environment**, which is the SimuLearn Web Platform when a user interacts with the system. This includes the actions like uploading solutions, entering skill data, or navigating the dashboard.

2. **Sensors -> Agent:** The **Sensors** detect the user actions. Then it will process raw inputs like a code file and text and turn them into **"Observations"**, which include specific data like user's skills, code solution, and progress. Then these data will be

sent to the Agent.

3. **Agent -> Actuators:** The **Agent** will receive these observations. Then it uses its internal logic and rules to process the data and make **"Decisions"**. These decisions determine what needs to happen next, like grading a submission, generating specific feedback, and deciding to unlock the next task.

4. **Actuators -> Environment:** The Agent sends commands based on its decisions to the **Actuators**. The actuators execute these commands and change the state in the **Environment**. For example, it will update the UI with the grade, display feedback and enable a new task button. This updated environment sets the stage for the user's next interaction then restart the cycle.

3.0 Explanation In Proof Of Concept (POC)

4.0 Behaviors Of Agent To Achieve The System Goals